AAE1001 Introduction to Artificial Intelligence and Data Analytics in Aerospace and Aviation Engineering

Week 9 (Project Additional Tasks)

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Additional Tasks (optional as bonus)

1. **Adding Checkpoints**
2. **Changing Environment**
3. To be released…

*Start working on the following Tasks after you finish the previous ones
(Create separate .py files so these tasks don't affect each other)

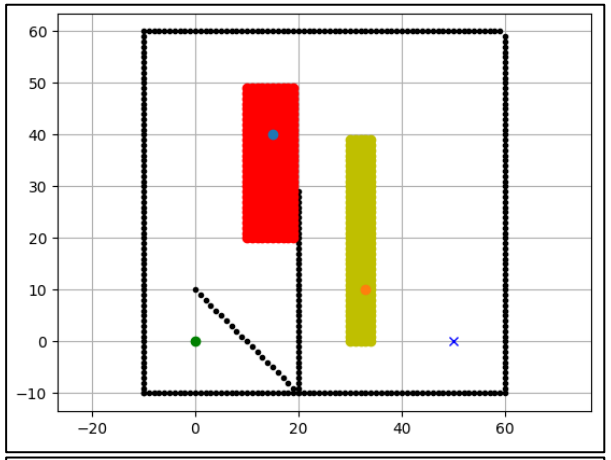
Try to incorporate GenAI to help you!

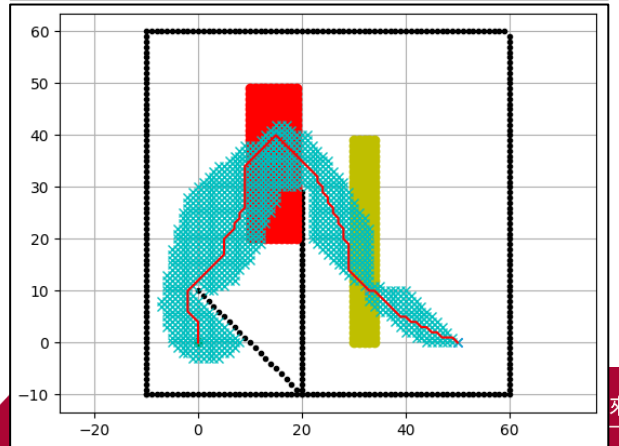
Task A1 - Adding a Checkpoint



Assume the aircraft is a supply craft that must reach 2 drop-off points to drop supplies before heading to base.

- Add one checkpoint for each cost intensive area (2 in total)
- Reach all checkpoints before arriving at the destination





Requirements:

1. This is an add-on for the code you are currently working on
2. Checkpoints should be generated inside the cost intensive areas
3. Plot the checkpoints together with your planned path with appropriate visualization

Task A2 - Changing Environment

Assume the mission and the environment keep changing for each operation.

Modify the code so that:

1. Only the fuel-consuming area remains and generate it randomly **with a fixed area (40x40)**
2. Diagonal movement is **disabled**, change parameter(s) so that the object could travel **within one grid size**
3. Obstacles are generated randomly with **reasonable density**
4. Destination and starting points are generated randomly with **at least a 40-unit distance in-between**
5. Plotting of the fuel-consuming area would not cover the obstacles, and obstacles **should not generate** at/near the start and end point

